Hidenori Tani

Ph.D. in Engineering

Associate Professor, Department of Health Pharmacy, Yokohama University of Pharmacy (2023–)
Faculty of Pharmaceutical Sciences, Yokohama University of Pharmacy
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ORCID | Google Scholar | GitHub | Publications | Data & Code

62	2,323	23	34
PUBLICATIONS	CITATIONS	H-INDEX	I10-INDEX

Source: Google Scholar (as of 2026-07-20); publication count from ORCID

RESEARCH SUMMARY

Molecular biologist specializing in **long noncoding RNAs (lncRNAs)** and their metabolism. Developer of **BRIC-seq**, a genome-wide method for measuring intracellular RNA stability (half-life), used to study RNA degradation as a facet of function. Current work extends to single-cell RNA turnover, RNA drug development, and computational/AI approaches to RNA biology.

RESEARCH INTERESTS

- Function and metabolism of long noncoding RNAs (lncRNAs)
- Genome-wide measurement of RNA stability / half-life (BRIC-seq)
- Post-transcriptional regulation and RNA degradation pathways
- Single-cell RNA turnover
- RNA drug development; computational and AI approaches to RNA biology

APPOINTMENTS

2023 – **Associate Professor**, Yokohama University of Pharmacy
横浜薬科大学 薬学部 健康薬学科

2012 – 2023 **Researcher → Senior Researcher**, AIST
産業技術総合研究所

2009 – 2012 **Postdoctoral Researcher**, University of Tokyo
東京大学

EDUCATION

2008 **Ph.D. in Engineering**, Waseda University
早稲田大学大学院 理工学研究科

B.Eng., Applied Chemistry, Waseda University
早稲田大学 理工学部 応用化学科

METHOD & CONTRIBUTION

Developed **BRIC-seq (5′-bromo-uridine immunoprecipitation chase sequencing)**, establishing genome-wide measurement of intracellular RNA stability (half-life) and identifying hundreds of short-lived noncoding transcripts (Genome Research, 2012). Subsequent work consistently treats RNA stability as an indicator of function.

SELECTED PUBLICATIONS

1. **Tani H.**, Mizutani R., Salam K.A., Tano K., Ijiri K., Wakamatsu A., et al. Genome-wide determination of RNA stability reveals hundreds of short-lived noncoding transcripts in mammals. Genome Research 22 (2012). doi:10.1101/gr.130559.111
2. **Tani H.**, Imamachi N., Salam K.A., Mizutani R., Ijiri K., Irie T., et al. Identification of hundreds of novel UPF1 target transcripts by direct determination of whole transcriptome stability. RNA Biology 9 (2012). doi:10.4161/rna.22360
3. **Tani H.**, Torimura M., Akimitsu N. The RNA Degradation Pathway Regulates the Function of GAS5 a Non-Coding RNA in Mammalian Cells. PLoS ONE 8 (2013). doi:10.1371/journal.pone.0055684
4. **Tani H.**, Akimitsu N. Genome-wide technology for determining RNA stability in mammalian cells. RNA Biology 9 (2012). doi:10.4161/rna.22036
5. Imamachi N., **Tani H.**, Mizutani R., Imamura K., Irie T., Suzuki Y., et al. BRIC-seq: A genome-wide approach for determining RNA stability in mammalian cells. Methods 67 (2014). doi:10.1016/j.ymeth.2013.07.014
6. **Tani H.**, Onuma Y., Ito Y., Torimura M. Long Non-Coding RNAs as Surrogate Indicators for Chemical Stress Responses in Human-Induced Pluripotent Stem Cells. PLoS ONE 9 (2014). doi:10.1371/journal.pone.0106282
7. **Tani H.**, Torimura M. Identification of short-lived long non-coding RNAs as surrogate indicators for chemical stress response. Biochemical and Biophysical Research Communications 439 (2013). doi:10.1016/j.bbrc.2013.09.006
8. **Tani H.** Recent Advances and Prospects in RNA Drug Development. International Journal of Molecular Sciences 25 (2024). doi:10.3390/ijms252212284
9. **Tani H.**, Teramura T., Adachi K., Tsuneda S., Kurata S., Nakamura K., et al. Technique for Quantitative Detection of Specific DNA Sequences Using Alternately Binding Quenching Probe Competitive Assay Combined with Loop-Mediated Isothermal Amplification. Analytical Chemistry 79 (2007). doi:10.1021/ac070041e
10. Mizutani R., Wakamatsu A., Tanaka N., Yoshida H., Tochigi N., Suzuki Y., et al. Identification and Characterization of Novel Genotoxic Stress-Inducible Nuclear Long Noncoding RNAs in Mammalian Cells. PLoS ONE 7 (2012). doi:10.1371/journal.pone.0034949
11. Maekawa S., Imamachi N., Irie T., **Tani H.**, Matsumoto K., Mizutani R., et al. Analysis of RNA decay factor mediated RNA stability contributions on RNA abundance. BMC Genomics 16 (2015). doi:10.1186/s12864-015-1358-y
12. **Tani H.**, Miyata R., Ichikawa K., Morishita S., Kurata S., Nakamura K., et al. Universal Quenching Probe System: Flexible, Specific, and Cost-Effective Real-Time Polymerase Chain Reaction Method. Analytical Chemistry 81 (2009). doi:10.1021/ac900414u

TEACHING

Courses taught, Yokohama University of Pharmacy

- Introduction to Information Science (1st year)
- Information Processing Practicum (1st year)
- Biological Pharmacy Seminar I (4th year)
- Biology Laboratory II — immunology (2nd year; course coordinator)

SERVICE & PEER REVIEW

- Reviewer: *Microbiology and Molecular Biology Reviews* (American Society for Microbiology), among others

BOOKS (GENERAL AUDIENCE)

Author of science and practical books for general readers, researchers, and graduate students. See the books page.

FULL PUBLICATION LIST

2026

Tani H. RNA-Binding Protein Occupancy Composition Predicts Long Noncoding RNA Subcellular Localization. *International Journal of Molecular Sciences* 27 (2026). doi:10.3390/ijms27125593

2025

Tani H. Biomolecules Interacting with Long Noncoding RNAs. *Biology* 14 (2025). doi:10.3390/biology14040442

Tani H. RMST: a long noncoding RNA involved in cancer and disease. *The Journal of Biochemistry* 177 (2025). doi:10.1093/jb/mvae083

Tani H. Biomolecules Interacting with Long Noncoding RNAs. (2025). doi:10.20944/preprints202504.0252.v1

2024

Tani H. Metabolic labeling of RNA using ribonucleoside analogs enables the evaluation of RNA synthesis and degradation rates. *Analytical Sciences* 41 (2024). doi:10.1007/s44211-024-00704-6

Tani H. Recent Advances and Prospects in RNA Drug Development. *International Journal of Molecular Sciences* 25 (2024). doi:10.3390/ijms252212284

Yokoyama S., Muto H., Honda T., Kurokawa Y., Ogawa H., Nakajima R., Kawashima H., **Tani H.** Identification of Two Long Noncoding RNAs, Kcnq1ot1 and Rmst, as Biomarkers in Chronic Liver Diseases in Mice. *International Journal of Molecular Sciences* 25 (2024). doi:10.3390/ijms25168927

Yagi Y., Abe R., **Tani H.** Exploring IDI2-AS1, OIP5-AS1, and LITATS1: Changes in Long Non-coding RNAs Induced by the Poly I:C Stimulation. *Biological and Pharmaceutical Bulletin* 47 (2024). doi:10.1248/bpb.b24-00037

2023

Abe R., Yagi Y., **Tani H.** Identifying Long Non-Coding RNA as Potential Indicators of Bacterial Stress in Human Cells. *BPB Reports* 6 (2023). doi:10.1248/bpbreports.6.6_226

2021

Tani H., Yamaguchi M., Enomoto Y., Matsumura Y., Habe H., Nakazato T., Kurata S. Naked-eye detection of specific DNA sequences amplified by the polymerase chain reaction with nanocomposite beads. *Analytical Biochemistry* 617 (2021). doi:10.1016/j.ab.2021.114114

2020

Aoki H., **Tani H.**, Nakamura K., Sato H., Torimura M., Nakazato T. MicroRNA biomarkers for chemical hazard screening identified by RNA deep sequencing analysis in mouse embryonic stem cells. *Toxicology and Applied Pharmacology* 392 (2020). doi:10.1016/j.taap.2020.114929

2019

Tani H., Numajiri A., Aoki M., Umemura T., Nakazato T. Short-lived long noncoding RNAs as surrogate indicators for chemical stress in HepG2 cells and their degradation by nuclear RNases. *Scientific Reports* 9 (2019). doi:10.1038/s41598-019-56869-y

TANI H. Development and Application of Analytical Methods for Biological Molecules Using the Fluorescent Dyes and the Nucleotide Analogs. *BUNSEKI KAGAKU* 68 (2019). doi:10.2116/bunsekikagaku.68.109

2018

Tani H., Matsutani T., Aoki H., Nakamura K., Hamaguchi Y., Nakazato T., Hamada M. Identification of RNA biomarkers for chemical safety screening in neural cells derived from mouse embryonic stem cells using RNA deep sequencing analysis. *Biochemical and Biophysical Research Communications* 512 (2018). doi:10.1016/j.bbrc.2018.11.141

Multiplicity in long noncoding RNA in living cells.. *Biomedical Sciences* (2018)

ノンコーディングRNAから生命の謎を解き明かす. (2018)

ヒトiPS細胞を用いた水質安全性評価のシステム. (2018)

蛍光色素及び修飾核酸を利用した生体分子解析技術の開発とその応用. *ぶんせき* (2018)

2017

Tani H., Takeshita J.i., Aoki H., Nakamura K., Abe R., Toyoda A., Endo Y., Miyamoto S., et al. Effect of methyl p-hydroxybenzoate on the culture of mammalian cell. *Drug Discoveries & Therapeutics* 11 (2017). doi:10.5582/ddt.2017.01054

Hermawan I., Furuta A., Higashi M., Fujita Y., Akimitsu N., Yamashita A., Moriishi K., Tsuneda S., et al. Four Aromatic Sulfates with an Inhibitory Effect against HCV NS3 Helicase from the Crinoid Alloeocomatella polycladia. *Marine Drugs* 15 (2017). doi:10.3390/md15040117

Tani H., Takeshita J.i., Aoki H., Nakamura K., Abe R., Toyoda A., Endo Y., Miyamoto S., et al. Identification of RNA biomarkers for chemical safety screening in mouse embryonic stem cells using RNA deep sequencing analysis. *PLOS ONE* 12 (2017). doi:10.1371/journal.pone.0182032

Tani H., Sato H., Torimura M. Rapid monitoring of RNA degradation activity in vivo for mammalian cells. *Journal of Bioscience and Bioengineering* 123 (2017). doi:10.1016/j.jbiosc.2016.11.010

Tani H., Okuda S., Nakamura K., Aoki M., Umemura T. Short-lived long non-coding RNAs as surrogate indicators for chemical exposure and LINC00152 and MALAT1 modulate their neighboring genes. *PLOS ONE* 12 (2017). doi:10.1371/journal.pone.0181628

Tani H. Short-lived non-coding transcripts (SLiTs): Clues to regulatory long non-coding RNA. *Drug Discoveries & Therapeutics* 11 (2017). doi:10.5582/ddt.2017.01002

ヒト細胞を用いた化学物質の安全性評価. (2017)

次世代シークエンサーによるノンコーディングRNAの解析. (2017)

2016

Tani H., Takeshita J.i., Aoki H., Abe R., Toyoda A., Endo Y., Miyamoto S., Gamo M., et al. Genome-wide gene expression analysis of mouse embryonic stem cells exposed to p-dichlorobenzene. *Journal of Bioscience and Bioengineering* 122 (2016). doi:10.1016/j.jbiosc.2016.02.007

ノンコーディングRNA動態解明の最前線. (2016)

2015

Furuta A., Salam K.A., **Tani H.**, Tsuneda S., Sekiguchi Y., Akimitsu N., Noda N. A Fluorescence-Based Screening Assay for Identification of Hepatitis C Virus NS3 Helicase Inhibitors and Characterization of Their Inhibitory Mechanism. *Methods in Molecular Biology* (2015). doi:10.1007/978-1-4939-2214-7_14

Maekawa S., Imamachi N., Irie T., **Tani H.**, Matsumoto K., Mizutani R., Imamura K., Kakeda M., et al. Analysis of RNA decay factor mediated RNA stability contributions on RNA abundance. *BMC Genomics* 16 (2015). doi:10.1186/s12864-015-1358-y

Tani H., Torimura M. Development of cytotoxicity-sensitive human cells using overexpression of long non-coding RNAs. *Journal of Bioscience and Bioengineering* 119 (2015). doi:10.1016/j.jbiosc.2014.10.012

Tani H., Imamachi N., Mizutani R., Imamura K., Kwon Y., Miyazaki S., Maekawa S., Suzuki Y., et al. Genome-Wide Analysis of Long Noncoding RNA Turnover. *Methods in Molecular Biology* (2015). doi:10.1007/978-1-4939-2253-6_19

Furuta A., Tsubuki M., Endoh M., Miyamoto T., Tanaka J., Salam K., Akimitsu N., **Tani H.**, et al. Identification of Hydroxyanthraquinones as Novel Inhibitors of Hepatitis C Virus NS3 Helicase. *International Journal of Molecular Sciences* 16 (2015). doi:10.3390/ijms160818439

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Furuta A., Salam K., Hermawan I., Akimitsu N., Tanaka J., **Tani H.**, Yamashita A., Moriishi K., et al. Identification and Biochemical Characterization of Halisulfate 3 and Suvanine as Novel Inhibitors of Hepatitis C Virus NS3 Helicase from a Marine Sponge. *Marine Drugs* 12 (2014). doi:10.3390/md12010462

Tani H., Onuma Y., Ito Y., Torimura M. Long Non-Coding RNAs as Surrogate Indicators for Chemical Stress Responses in Human-Induced Pluripotent Stem Cells. *PLoS ONE* 9 (2014). doi:10.1371/journal.pone.0106282

Salam K., Furuta A., Noda N., Tsuneda S., Sekiguchi Y., Yamashita A., Moriishi K., Nakakoshi M., et al. PBDE: Structure-Activity Studies for the Inhibition of Hepatitis C Virus NS3 Helicase. *Molecules* 19 (2014). doi:10.3390/molecules19044006

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Tani H., Mizutani R., Salam K.A., Tano K., Ijiri K., Wakamatsu A., Isogai T., Suzuki Y., et al. Genome-wide determination of RNA stability reveals hundreds of short-lived noncoding transcripts in mammals. *Genome Research* 22 (2012). doi:10.1101/gr.130559.111

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Mizutani R., Wakamatsu A., Tanaka N., Yoshida H., Tochigi N., Suzuki Y., Oonishi T., **Tani H.**, et al. Identification and Characterization of Novel Genotoxic Stress-Inducible Nuclear Long Noncoding RNAs in Mammalian Cells. *PLoS ONE* 7 (2012). doi:10.1371/journal.pone.0034949

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Fujimoto Y., Salam K.A., Furuta A., Matsuda Y., Fujita O., **Tani H.**, Ikeda M., Kato N., et al. Inhibition of Both Protease and Helicase Activities of Hepatitis C Virus NS3 by an Ethyl Acetate Extract of Marine Sponge Amphimedon sp. *PLoS ONE* 7 (2012). doi:10.1371/journal.pone.0048685

Salam K.A., Furuta A., Noda N., Tsuneda S., Sekiguchi Y., Yamashita A., Moriishi K., Nakakoshi M., et al. Inhibition of Hepatitis C Virus NS3 Helicase by Manoalide. *Journal of Natural Products* 75 (2012). doi:10.1021/np200883s

Yamashita A., Salam K.A., Furuta A., Matsuda Y., Fujita O., **Tani H.**, Fujita Y., Fujimoto Y., et al. Inhibition of Hepatitis C Virus Replication and Viral Helicase by Ethyl Acetate Extract of the Marine Feather Star Alloeocomatella polycladia. *Marine Drugs* 10 (2012). doi:10.3390/md10040744

RNAの安定性を指標にして機能不明な長鎖ノンコーディングRNAの性質を定義する. (2012)

Up-frameshift protein 1 (UPF1): multitalented entertainer in RNA decay. *Drug discoveries & therapeutics* (2012)

2010

Miyata R., Adachi K., **Tani H.**, Kurata S., Nakamura K., Tsuneda S., Sekiguchi Y., Noda N. Quantitative detection of chloroethene-reductive bacteria Dehalococcoides spp. using alternately binding probe competitive polymerase chain reaction. *Molecular and Cellular Probes* 24 (2010). doi:10.1016/j.mcp.2009.11.005

Tani H., Fujita O., Furuta A., Matsuda Y., Miyata R., Akimitsu N., Tanaka J., Tsuneda S., et al. Real-time monitoring of RNA helicase activity using fluorescence resonance energy transfer in vitro. *Biochemical and Biophysical Research Communications* 393 (2010). doi:10.1016/j.bbrc.2010.01.100

Stability of MALAT-1, a nuclear long non-coding RNA in mammalian cells, varies in various cancer cells. *Drug discoveries & therapeutics* (2010)

2009

Tani H., Akimitsu N., Fujita O., Matsuda Y., Miyata R., Tsuneda S., Igarashi M., Sekiguchi Y., et al. High-throughput screening assay of hepatitis C virus helicase inhibitors using fluorescence-quenching phenomenon. *Biochemical and Biophysical Research Communications* 379 (2009). doi:10.1016/j.bbrc.2009.01.020

Morishita S., **Tani H.**, Kurata S., Nakamura K., Tsuneda S., Sekiguchi Y., Noda N. Real-time reverse transcription loop-mediated isothermal amplification for rapid and simple quantification of WT1 mRNA. *Clinical Biochemistry* 42 (2009). doi:10.1016/j.clinbiochem.2009.01.013

Tani H., Miyata R., Ichikawa K., Morishita S., Kurata S., Nakamura K., Tsuneda S., Sekiguchi Y., et al. Universal Quenching Probe System: Flexible, Specific, and Cost-Effective Real-Time Polymerase Chain Reaction Method. *Analytical Chemistry* 81 (2009). doi:10.1021/ac900414u

メタゲノム解析におけるFunction-based screening法の高度化. *マリンメタゲノムの有効利用* (2009)

2008

Noda N., **Tani H.**, Morita N., Kurata S., Nakamura K., Kanagawa T., Tsuneda S., Sekiguchi Y. Estimation of single-nucleotide polymorphism allele frequency by alternately binding probe competitive polymerase chain reaction. *Analytica Chimica Acta* 608 (2008). doi:10.1016/j.aca.2007.12.015

2007

Tani H., Kanagawa T., Morita N., Kurata S., Nakamura K., Tsuneda S., Noda N. Calibration-curve-free quantitative PCR: A quantitative method for specific nucleic acid sequences without using calibration curves. *Analytical Biochemistry* 369 (2007). doi:10.1016/j.ab.2007.06.047

Tani H., Kanagawa T., Kurata S., Teramura T., Nakamura K., Tsuneda S., Noda N. Quantitative Method for Specific Nucleic Acid Sequences Using Competitive Polymerase Chain Reaction with an Alternately Binding Probe. *Analytical Chemistry* 79 (2007). doi:10.1021/ac061506o

Tani H., Teramura T., Adachi K., Tsuneda S., Kurata S., Nakamura K., Kanagawa T., Noda N. Technique for Quantitative Detection of Specific DNA Sequences Using Alternately Binding Quenching Probe Competitive Assay Combined with Loop-Mediated Isothermal Amplification. *Analytical Chemistry* 79 (2007). doi:10.1021/ac070041e

2005

Tani H., Noda N., Yamada K., Kurata S., Tsuneda S., Hirata A., Kanagawa T. Quantification of Genetically Modified Soybean by Quenching Probe Polymerase Chain Reaction. *Journal of Agricultural and Food Chemistry* 53 (2005). doi:10.1021/jf048031r

Tani in bold. The complete, always-current list is on the publications page / ORCID.

As of 2026-07-20